

Karim ElSedfy

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EDUCATION

Bowdoin College

B.A. in Computer Science and Mathematics

May 2028

Brunswick, ME

- **GPA:** 3.85/4.00, Dean's List
- **Relevant Coursework:** Deep Learning, Artificial Intelligence, Algorithms, Computer Systems, Data Structures I & II, Linear Algebra

EXPERIENCE

Dropbox

Software Engineer Intern

May 2026 – Present

Remote - San Francisco, CA

- Ship full-stack product changes across a React/TypeScript frontend and Python backend services, using internal gRPC RPC and A/B experimentation tooling to launch and measure targeted conversion flows.
- Build onboarding and upgrade experiences on the Teams Formation team that move business-domain users on free plans toward paid Team plans, tying engineering changes to adoption and recurring revenue.

Palmera Systems Co.

Software Engineer Intern

Jun. 2025 – Aug. 2025

Cairo, Egypt

- Architected face-recognition and anti-spoofing for the smartIN app (8K daily users), leading dataset curation and model fine-tuning with DeepFace/OpenCV to reach 96% detection accuracy.
- Built responsive web frontends and backend REST services on PostgreSQL matching live captures in under 2s, cutting authentication latency by 50%.

Un gated Research, Bowdoin College

Software Engineer

Mar. 2026 – Present

Brunswick, ME

- Co-maintain a multithreaded Python ETL pipeline (SQLAlchemy, MariaDB, AWS S3) ingesting economics papers from 11+ journals, enriched with LLM-generated summaries, tags, and translations.
- Built error-detection and automated-repair diagnostics, and ran Crossref API polling across 10 years of publications to surface 6,080 new articles for ingestion.

Computer Science Department, Bowdoin College

Independent Study Researcher

Sep. 2025 – Dec. 2025

Brunswick, ME

- Researched transferring detection-trained YOLO encoders to semantic segmentation without retraining the backbone, using dense stride-1 inference, frozen-encoder decoders, and U-Net-style skip connections.
- Improved mean IoU from 0.600 to 0.684 on the CUB-200 dataset and wrote the results as an IEEE-format paper under Prof. Jeova Farias.

PROJECTS

Persona-Motion: Audio-Driven Talking-Head Synthesis

[GitHub](#)

Python, PyTorch, Flow Matching, LivePortrait, Slurm HPC

- Built an audio-driven talking-head generator: a 14.7M-parameter rectified flow-matching model maps 768-d pretrained speech-encoder features to 205-d facial-motion coefficients at 20 fps, rendered to video by a frozen LivePortrait animator.
- Ran the full pipeline on a Slurm GPU cluster (~6.4 h of GPU motion extraction per 16-clip dataset; 300-epoch training runs in ~22 min), improving lip-sync ~6× over an MSE regression baseline (sync-confidence 0.09 → 0.56 vs. 0.77 ground truth) with classifier-free guidance, evaluated on a from-scratch contrastive SyncNet metric.

Proteus: Virtual Try-On

[Frontend](#) | [Backend](#) | [Live Demo](#)

React, Node.js, Express, Redis, PostgreSQL, Python, PyTorch, Kubernetes, KEDA, S3

- Built an event-driven, GPU-accelerated virtual try-on system where a Redis queue and pub/sub feed a Python/PyTorch worker running CatVTON on Linode Kubernetes Engine.
- Delivered real-time product UX with WebSocket completion events, presigned S3 uploads, and backend job orchestration that lets users upload a photo and receive synthesized try-on output without page reloads.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C#

AI / Data: PyTorch, generative modeling (flow matching, CFG), OpenCV, Hugging Face Transformers, SQLAlchemy, LLM-powered data pipelines

Frameworks and Libraries: React, Node.js, Express.js, FastAPI, Flask

Technologies and Tools: PostgreSQL, MariaDB, Redis, Docker, Kubernetes, Slurm (HPC), AWS S3, GCP, Firebase, gRPC, Git